

EMILY LIANG

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Education

The University of Texas at Austin

May 2028 (Expected)

B.S. Computer Science (Turing Scholars Honors); B.S. Mathematics; Minor in Business

GPA: 3.86

Relevant Coursework: Programming Languages, Artificial Intelligence, Algorithms, Machine Learning, Operating Systems, Computer Architecture, Compilers, Computer Vision, Probability, Stochastic Processes, Number Theory

Experience

Quantitative Developer Intern

May 2026 – Aug. 2026

Base Power Company

Austin, TX

- Developed an optimization-based battery dispatch algorithm under ERCOT market and operational constraints, improving expected profitability from \$1/kWh to \$3/kWh.
- Built simulation and cross-validation infrastructure for battery dispatch strategies across historical ERCOT market conditions, reducing backtest error by 15%.
- Quantified the impact of price forecasts, battery degradation, and operational constraints on dispatch profitability to inform objective functions and optimization parameters.

Undergraduate Researcher

[Start Date] – Present

Learning Directed Operating Systems (LDOS), The University of Texas at Austin

Austin, TX

- Develop robotic-system simulations in NVIDIA Isaac Sim to evaluate robot behavior under varying hardware and system configurations.
- Implement hardware-aware simulation models for GPU and CPU computation and network communication delays to model end-to-end robotic system performance.
- Analyze the impact of compute and networking constraints on robotics workloads to characterize system bottlenecks and hardware-performance tradeoffs.

Simulation and Validation System Lead

Aug. 2024 – Present

Longhorn Racing Internal Combustion

Austin, TX

- Lead a team of 7 engineers developing simulation, telemetry, and validation infrastructure for a Formula SAE race car, setting technical priorities and coordinating integration across vehicle systems.
- Develop a quasi-steady-state lap-time simulator integrating GGV performance envelopes, track trajectories, and configurable vehicle models to predict feasible velocity profiles and lap times.
- Direct track-validation experiments and parameter studies to calibrate simulation models, quantify sensitivity to vehicle design variables, and inform engineering design decisions.

Undergraduate Course Assistant

Aug. 2026 – Present

The University of Texas at Austin

Austin, TX

- Mentor students in Software Engineering and Freshman Think Lab through office hours, code and design reviews, and technical guidance on software architecture, debugging, testing, and version control.

Projects

JPEB: Custom 16-Bit Computer | *Rust, Haskell, C, Verilog, Python, Assembly*

- Designed and implemented a custom 16-bit computer in a team of four, spanning ISA design, processor architecture, compiler tooling, and emulation.
- Developed a 6-stage pipelined processor in Verilog and deployed it to an FPGA, implementing hazard detection, data forwarding, pipeline stalls, and control-flow flushing.
- Extended the compiler and assembler toolchain and implemented Chrome Dinosaur and Snake using the custom ISA to validate end-to-end hardware and software execution.

LiveOak Compiler | *OCaml, x86-64 Assembly*

- Built a compiler in OCaml for a statically typed programming language, implementing lexical analysis, parsing, semantic analysis, and x86-64 code generation.
- Designed AST transformations and type-checking passes to validate program semantics and lower high-level language constructs into intermediate representations.
- Implemented backend code generation for control flow, function calls, expressions, and memory operations, producing executable x86-64 assembly.

x86 Multicore Operating System Kernel | *C++*

- Built a preemptive multicore x86 operating system kernel supporting concurrent process and thread execution with synchronization primitives.
- Implemented virtual memory with demand paging, page caching, swapping, and memory-mapped files.
- Developed kernel-level process, thread, file-system, and memory-management system calls.

Technical Skills

Languages: C/C++, Python, Go, Java, JavaScript/TypeScript, Verilog, x86 Assembly, ARM Assembly, OCaml

Tools/Frameworks: Git, Docker, Linux, NumPy, SciPy, Pandas, Matplotlib, PyTorch